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Design and construction of a Pound-Drever-Hall locking setup for laser frequency stabilization to a reference multi-bore optical cavity

Semester thesis

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Abstract

Laser frequency stabilization is an essential requirement for trapped-ion experiments, enabling coherent control and high-precision measurements. Methods for achieving stable laser frequencies include the Pound-Drever-Hall (PDH) locking technique, which stabilizes the laser frequency to an optical cavity. While PDH locking is a well-established technique, building a functional setup requires careful alignment of optics, cavity mode matching, and electronic feedback control. In this work, we designed and implemented a PDH locking scheme to stabilize four lasers to a reference multi-bore cavity. We tested and characterized the frequency locks, demonstrating high mode-matching efficiency, consistent locking performance, and robust response to disturbances. The successful stabilization of multiple laser frequencies marks an important step toward conducting high-precision experiments with trapped ions.

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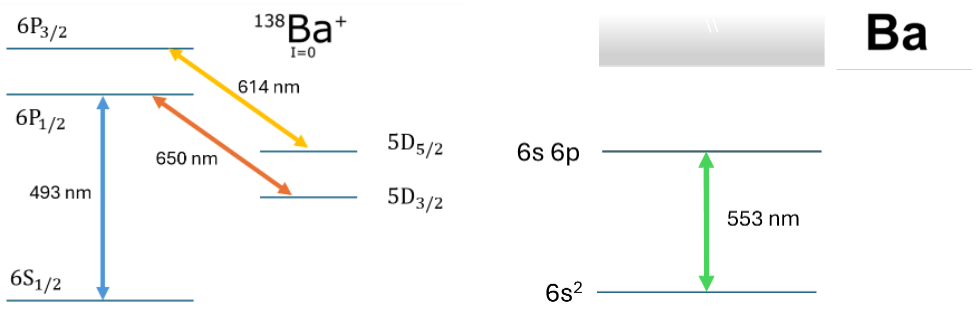
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Chapter 1

Introduction

Trapped-ion systems provide a controllable platform not only for quantum information processing, but also for precision measurements and tests of fundamental physics. The present work is part of a broader effort to measure atomic parity violation (APV) in a trapped Barium ion (Ba+) system [1].

Several laser wavelengths are required in the Ba+ setup (see Fig. 1.1). Among others, the 493 nm laser is used for Doppler cooling and state detection, forming the basis for state preparation and fluorescence readout. The 650 nm and 614 nm lasers act as repumpers to maintain a closed cooling cycle. A 553 nm laser is employed for the photoionization of neutral barium during ion preparation. In our lab, these wavelengths are generated using different



(a) Barium ion electronic levels **(b)** Neutral Barium electronic levels

Figure 1.1: Electronic energy level diagrams for the $^{138}Ba^{+}$ ion and neutral Barium. (a) Level scheme of the $^{138}Ba^{+}$ ion showing the 493 nm cooling transition and the 614/650 nm repumpers. (b) Relevant electronic levels for neutral Barium, highlighting the 553 nm transition used for the photoionization process. The grey zone is where the unbound electrons are located: to ionize Barium when the valence electron is in 6s6p, it's sufficient to use any laser with a frequency higher than 6s6p binding energy.

laser systems. Infrared light at 1228 and 1107 nm is produced by TOPTICA

MTA Pro laser system, consisting of a diode laser seed followed by a tapered amplifier for power amplification. These infrared beams are subsequently frequency-doubled to obtain 614 nm and 553 nm, respectively. The 493 nm and 650 nm beams are provided by TOPTICA MDL Pro diode lasers.

All laser sources are susceptible to frequency noise caused by various mechanisms, including current and temperature fluctuations in the diode lasers, mechanical vibrations in the laser cavity, acoustic noise, and environmental thermal drifts. Frequency fluctuations translate into detuning errors and phase noise, which degrade cooling efficiency, state detection fidelity, and coherence.

To achieve the required stability, we lock the lasers to an optical reference cavity using the Pound–Drever–Hall (PDH) technique. PDH locking provides an error signal proportional to frequency detuning, enabling active feedback control with high robustness to perturbations. Although conceptually well established, the implementation of a reliable PDH system requires careful optical alignment and optimized electronic feedback.

In this thesis, we present the design of a PDH-based frequency stabilization system for four lasers referenced to a multi-bore cavity and the implementation of the setup for 493 nm and 650 nm wavelengths. The successful stabilization of the 493 nm and 650 nm lasers represents a key technical step toward high-precision APV measurements in trapped Ba+ ions.

Chapter 2

Theoretical Background

2.1 Fundamentals of Optics

In this chapter, we introduce the theoretical foundations necessary to understand and implement a Pound-Drever-Hall setup. We start with a general overview on Gaussian beams optics and optical Fabry - Pérot cavities. Then, we introduce the working principles of laser frequency stabilization with a PDH lock system. Lastly, we provide basic concepts about feedback control systems in relation to a PDH lock. The theoretical framework for optical beam propagation and optical cavities theory presented in this work is based on standard references in classical optics and laser physics, such as Born and Wolf ([2]) for wave-optics formalism and Kogelnik ([3]) and Siegman ([4]) for the treatment of cavity resonances.

2.1.1 Gaussian Beams

The propagation of a monochromatic optical field in free space is described by the Helmholtz equation

$$\nabla^2 E(\mathbf{r}) + k^2 E(\mathbf{r}) = 0, \tag{2.1}$$

with $k = 2\pi/\lambda$ the wave vector, λ the wavelength, and $E(\mathbf{r})$ the position dependent electric field.

For beams predominantly propagating along the z-axis, we write

$$E(\mathbf{r}) = u(x, y, z)\, e^{ikz}, \tag{2.2}$$

where $u(x, y, z)$ is a slowly varying envelope and the remaining part is a plane wave propagation term. Under this paraxial approximation, we obtain the paraxial Helmholtz equation ([2]):

$$\nabla_\perp^2 u + 2ik\frac{\partial u}{\partial z} = 0, \tag{2.3}$$

with $\nabla_\perp^2$ the transverse Laplacian.

Its fundamental solution is the Gaussian beam. In practical optical systems, we describe laser fields well using Gaussian beams (Eq. 2.2) instead of plane waves. We characterize them by a Gaussian profile of the electric field intensity in the transverse plane. We usually describe it by the following parameters:

- Transverse size of the beam:

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2} \tag{2.4}$$

  where w_0 is the beam waist (minimum transverse size along the z-direction), and z_R is the Rayleigh range.

- Radius $R(z)$: we define the radius of curvature of the beam wavefront at position z.

- The complex beam parameter $q(z)$ provides a compact description of Gaussian beam propagation and is defined as

$$\frac{1}{q(z)} = \frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)}. \tag{2.5}$$

Under free-space propagation or interaction with optical elements, the q parameter transforms according to ABCD matrix formalism, making it a convenient tool to analyze beam evolution in optical systems:

$$q_{\text{out}} = \frac{A q_{\text{in}} + B}{C q_{\text{in}} + D}, \tag{2.6}$$

where A, B, C, D are the elements of the ABCD (or ray-transfer) matrix of the considered optical element.

In general, we can decompose any solution of the paraxial Helmholtz equation (2.3) into orthogonal Hermite-Gaussian modes in cartesian coordinates, or Laguerre-Gaussian modes in cylindrical coordinates ([4]). Considering Hermite-Gaussian modes, we index them by two integers (l, m) and note that they differ substantially in their spatial profile in the transverse plane (Fig. 2.1), but they present the same underlying Gaussian beam parameter structure (transverse beam size $w(z)$, Rayleigh range z_R, and beamfront radius $R(z)$). Thus, we can see any beam in an optical system as a superposition of Hermite-Gaussian modes. In the following, we adopt the standard notation TEM_{lm} (Transverse Electromagnetic) to label transverse modes with indices (l, m).

In addition to its amplitude profile, any Hermite-Gaussian mode exhibits a spatially varying phase. We define the total phase at position z along the

propagation axis as the sum of the usual plane-wave phase and the *Gouy phase* $\psi(z)$, which depends on the specific Hermite-Gaussian mode.

$$\Phi(x, y, z) = kz - \psi(z) + \frac{k(x^2 + y^2)}{2R(z)}, \tag{2.7}$$

where $R(z)$ is the radius of curvature of the wavefront, and

$$\psi(z) = (m + l + 1) \arctan\left(\frac{z}{z_R}\right) \tag{2.8}$$

is the Gouy phase. We account for the wavefront bending due to the radius of curvature $R(z)$ with the third term.

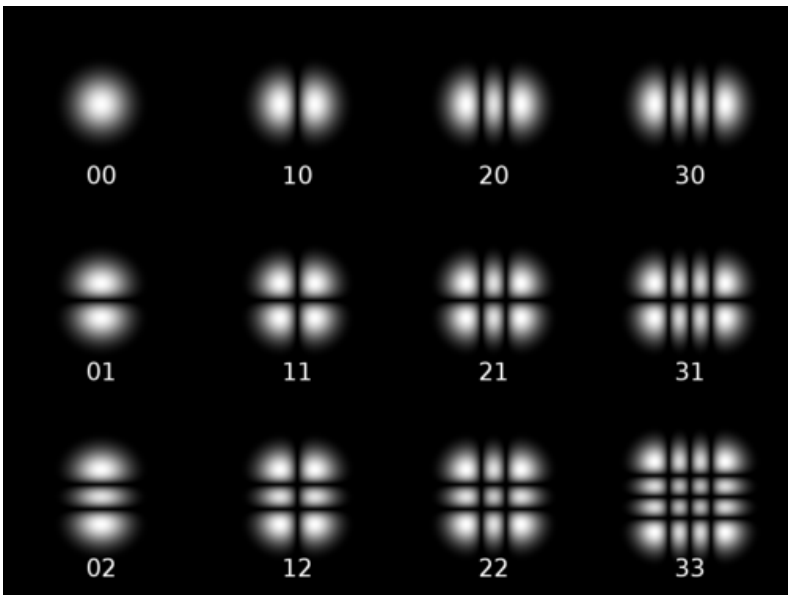


Figure 2.1: Spatial mode structure of the first twelve Hermite-Gaussian modes (reproduced from [5]). The indices represent the transverse mode numbers (l, m).

2.1.2 Optical cavities

An optical cavity (resonator) is an optical system that confines light in specific resonating modes. [3]. The most common optical cavity is made by two opposing reflecting mirrors aligned along the same axis, forming a Fabry - Perot cavity or etalon. This type of cavity is described by the length of the cavity L, the radius of curvature of the two mirrors R_1 and R_2 and by the reflection coefficients of the mirrors, defined as:

$$r = \frac{E_{\text{ref}}}{E_{\text{inc}}}, \tag{2.9}$$

where E_{inc} and E_{ref} are the incident electric field and the reflected electric field from the mirror.

In order to be resonant to the cavity, the reflections of the mirrors have to interfere constructively. This fact generates specific conditions for both the spatial profile of the resonant modes and their phase. The first condition forces the wave in the cavity, described within the Gaussian beam formalism, to have a radius of curvature equal to the radius of the mirrors, at the mirror surfaces.

The second condition on the phase requires a resonating mode to be in phase with itself after a round trip. This gives rise to a determined frequency spacing between two resonances (called longitudinal modes), referred as the free spectral range (FSR):

$$\text{FSR} = \frac{c}{2L}, \tag{2.10}$$

where L is the cavity length and c the speed of light.

Furthermore, Eq. 2.7 shows that each Hermite-Gaussian transverse mode TEM_{lm} satisfies a distinct phase relation, resulting in a different resonance frequency within a given free spectral range. As a consequence, the optical cavity supports multiple resonant modes at different angular frequencies ω_{nml}, corresponding to different spatial field (Hermite-Gaussian) distributions [6].

$$\frac{2\pi\omega_{nlm}}{\text{FSR}} = n + \frac{l+m+1}{\pi}\cos^{-1}\sqrt{\left(1-\frac{L}{R_1}\right)\left(1-\frac{L}{R_2}\right)}, \tag{2.11}$$

where n is the longitudinal integer mode number and R_1 and R_2 the radius of the cavity mirrors.

The amplitude of the total reflected electric field is also affected by the resonance frequencies in the cavity. From the expression of the interference between the reflected incoming field and the field leaking out from the first mirror, we can demonstrate that [7]:

$$F_{lm}(\omega) = \frac{E_{\text{ref}}^{\text{tot}}}{E_{\text{inc}}} = \frac{r(e^{i\varphi_{nlm}}-1)}{1-r^2e^{i\varphi_{nlm}}}, \tag{2.12}$$

where

$$\varphi_{lm} = \frac{\omega}{\text{FSR}} + \frac{l+m+1}{\pi}\cos^{-1}\sqrt{\left(1-\frac{L}{R_1}\right)\left(1-\frac{L}{R_2}\right)}.$$

Here we assumed that the cavity is made up of two lossless mirrors, with the same reflectivity. Taking the value of the reflected intensity:

$$I_{\text{refl}} = |F_{lm}(\omega)|^2 \cdot I_0 = I_0 \frac{\sin^2\left(\frac{\varphi_{lm}(\omega)}{2}\right)}{\left(\frac{2\mathcal{F}}{\pi}\right)^2 + \sin^2\left(\frac{\varphi_{lm}(\omega)}{2}\right)}, \tag{2.13}$$

where I_0 is the intensity of the incoming beam, and we introduced the Finesse of the cavity as:

$$\mathcal{F} = \frac{\pi\sqrt{R}}{1-R}, \tag{2.14}$$

with $R = |r|^2$.

The reflection coefficient $F(\omega)$ exhibits minima whenever the phase accumulated over one round trip satisfies the resonance condition, at frequencies given exactly by Eq. (2.11). For a cavity supporting multiple transverse modes, this condition is fulfilled at the different frequencies ω_{nlm}. Consequently, the reflected intensity $|F(\omega)|^2$ displays multiple dips within each free spectral range, each corresponding to the resonance of a distinct Hermite–Gaussian mode. The same pattern of reflection dips repeats itself every free spectral range. In Fig. 2.2, we plotted the transmission ($T_{lm} = 1 - |F_{lm}(\omega)|^2$, for the first four transverse Hermite-Gaussian modes.

Near resonance, the transmission peaks (or reflection dips) can be approximated in the following way:

$$T_{lm}(\omega) = \frac{I_0}{1 + \left(\frac{\mathcal{F}}{\pi \text{FSR}}\right)^2 (\omega - \omega_0)^2}, \tag{2.15}$$

where ω_0 is the considered resonance frequency. We identify Eq. 2.15 as a Lorentzian curve, with Full Width at Half Maximum (FWHM) in frequency units (Hz):

$$\delta\nu = \frac{\text{FSR}}{\mathcal{F}}. \tag{2.16}$$

Thus, we can also interpret the finesse of the cavity as the ratio between the free spectral range and the FWHM linewidth $\delta\nu$ of the resonance peaks of the cavity reflectance. As a result, a cavity with high finesse exhibits sharper peaks in the transmission spectrum.

2.2 Laser Frequency Stabilization

All lasers exhibit frequency fluctuations: any variation of the laser cavity length or of the refractive index of the cavity medium results in a corresponding shift of the laser frequency [4]. We classify these variations into long-term drifts, caused by temperature changes or ambient pressure fluctuations, and

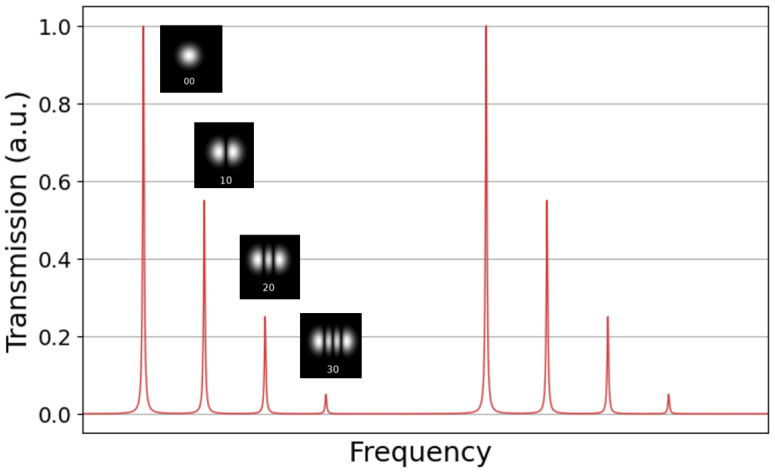


Figure 2.2: Simulated Fabry–Pérot transmission spectrum over two free spectral ranges (FSRs), showing four longitudinal cavity modes per FSR.

short-term drifts, originating from acoustic vibrations of the cavity mirrors or electrical noise in the laser driving electronics.

We use frequency stabilization techniques to reduce the effect of these fluctuations. In such schemes, we continuously measure the laser frequency with respect to a stable frequency reference and feed the resulting error signal back to a frequency-tuning element, such as a piezoelectric actuator applied to one of the laser cavity mirrors. We show the frequency stabilization feedback loop in Fig. 2.3.

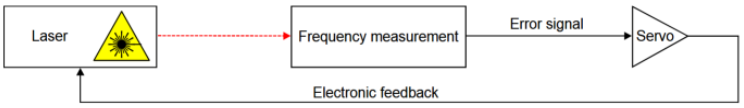


Figure 2.3: We describe the feedback loop for laser frequency stabilization. We measure the frequency with respect to a stable reference. We then generate an error signal, which carries information about the frequency deviation from the reference. We pass this information to an electronic servo, which corrects the deviation accordingly.

2.2.1 The Pound-Drever-Hall (PDH) Frequency Lock

A crucial part in laser frequency stabilization is to find a stable frequency reference system and identify a suitable error signal. The Pound-Drever-Hall technique [8] uses a Fabry-Perot cavity (as in Sec. 2.1.2) as a stable frequency reference. The key idea is to compare the laser frequency to an optical cavity by sending the laser beam into the cavity while operating at a well-defined point of the cavity transmission spectrum in frequency, and to monitor the

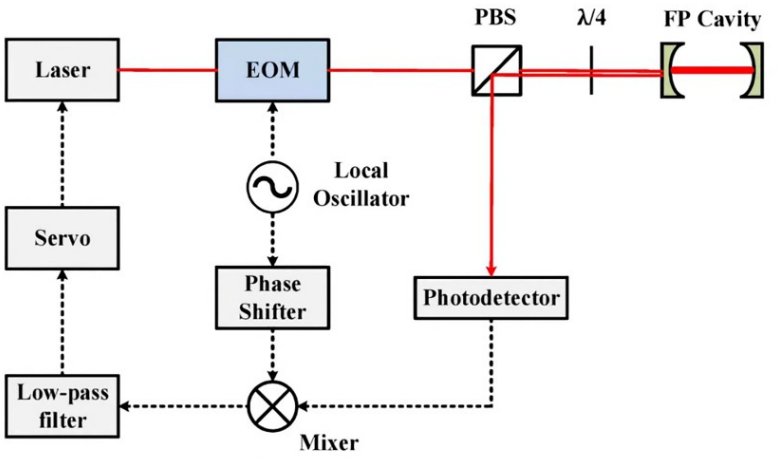


Figure 2.4: We describe the basic scheme for a PDH lock. The solid lines represent optical paths, while the dashed lines represent electronic signals. We phase modulate the laser beam using a EOM. We then detect the beam reflected from the cavity with a photodiode, and we mix the electronic signal with the modulation signal, followed by low-pass filtering to isolate the part carrying phase information. We include a phase shifter before the mixer to achieve full control over the error signal phase term $\cos\varphi$. Finally, we pass the obtained error signal to a servo that feeds it back to the laser [7].

reflected intensity. If the laser frequency is centered around a transmission peak (see Fig. 2.2), we can stabilize the laser frequency by keeping the reflected intensity at zero. However, this intensity-based error signal does not provide information about the sign of the detuning, meaning we cannot distinguish whether we need to increase or decrease the frequency to bring it back to the central resonance of the peak.

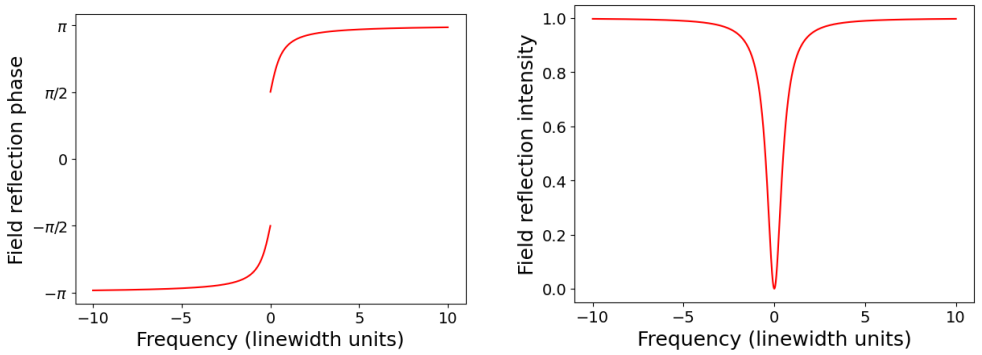
Instead, we access the phase of the total reflected electric field from the cavity, which is odd with respect to frequency detuning around resonance and therefore carries sign information (Fig. 2.5). The Pound–Drever–Hall technique exploits this fact by encoding frequency fluctuations into phase fluctuations of the reflected electric field through phase modulation of the incident laser.

In the following paragraphs, we show with calculations how phase modulation makes it possible to generate an effective error signal. A more detailed derivation can be also found in [7]. The incident electric field is phase-modulated at angular frequency Ω,

$$E_{\text{in}}(t) = E_0 e^{i\omega t + i\beta \sin(\Omega t)}, \tag{2.17}$$

where β is the modulation index. Using the Jacobi–Anger expansion (see [9]) and keeping only first-order terms, the field can be approximated as

$$E_{\text{in}}(t) \simeq E_0 \Big[J_0(\beta) e^{i\omega t} + J_1(\beta) e^{i(\omega+\Omega)t} - J_1(\beta) e^{i(\omega-\Omega)t} \Big], \tag{2.18}$$



(a) Reflected field phase

(b) Reflected field intensity

Figure 2.5: (a) Phase of the total reflected field $F(\omega)$, varying the laser frequency around one cavity transmission resonance. (b) Intensity of the total reflected field $|F(\omega)|^2$, around one cavity transmission resonance.

where J_0 and J_1 are the zero-th and first order Bessel functions. The field thus consists of a carrier at frequency ω and two first-order sidebands at $\omega \pm \Omega$.

After reflection from the cavity, each frequency component acquires the corresponding complex reflection coefficient $F(\omega)$ (Eq. 2.12),

$$E_{\text{ref}}(t) = E_0\Big[J_0 F(\omega)e^{i\omega t} + J_1 F(\omega+\Omega)e^{i(\omega+\Omega)t} - J_1 F(\omega-\Omega)e^{i(\omega-\Omega)t}\Big]. \quad (2.19)$$

The reflected light is detected by a photodiode, which measures the optical power:

$$P(t) \propto |E_{\text{ref}}(t)|^2. \quad (2.20)$$

Among the various contributions (constant terms and fast oscillating terms), we identify the terms oscillating at the modulation frequency Ω as arising from the interference between the carrier and the sidebands, and we can write them as

$$P(t) \supset 2\sqrt{P_c P_s}\Big[\Re\{\mathcal{K}(\omega)\}\cos(\Omega t) + \Im\{\mathcal{K}(\omega)\}\sin(\Omega t)\Big], \quad (2.21)$$

where

$$\mathcal{K}(\omega) = F(\omega)F^*(\omega+\Omega) - F^*(\omega)F(\omega-\Omega), \quad (2.22)$$

and P_c and P_s denote the optical powers in the carrier and sidebands, respectively (symbol $\supset$ means "contains"). This term contains the reflected phase information that we want.

If we choose the modulation frequency much larger than the cavity linewidth, the sidebands are fully reflected,

$$F(\omega \pm \Omega) \simeq -1. \quad (2.23)$$

In this limit,

$$\mathcal{K}(\omega) \simeq -2i\,\Im\{F(\omega)\}, \tag{2.24}$$

which is purely imaginary. Consequently, only the $\sin(\Omega t)$ term contributes to the signal.

To extract this component, we demodulate the photodiode signal by mixing it with $\sin(\Omega t + \varphi)$ and low-pass filtering. The resulting DC signal is:

$$\varepsilon_{\mathrm{PDH}}(\omega) \propto \Im\{F(\omega)\}\cos\varphi. \tag{2.25}$$

Choosing the demodulation phase $\varphi = 0$, the signal becomes directly proportional to the imaginary part of the cavity reflection coefficient,

$$\varepsilon_{\mathrm{PDH}}(\omega) \propto \Im\{F(\omega)\}. \tag{2.26}$$

Near resonance, $\Im\{F(\omega)\}$ varies linearly with frequency detuning ($\varepsilon_{PDH} \sim \frac{\delta\omega}{\delta\nu}$, where $\delta\nu$ is the cavity linewidth), providing a sign-sensitive and linear error signal suitable for laser frequency stabilization [7]. Moreover, most of the times it is useful to maximize the slope of the central part of the error signal to improve the feedback stability. It's possible to show [7] that this condition is achieved for a sideband to carrier power ratio $\frac{P_s}{P_c} = 0.42$.

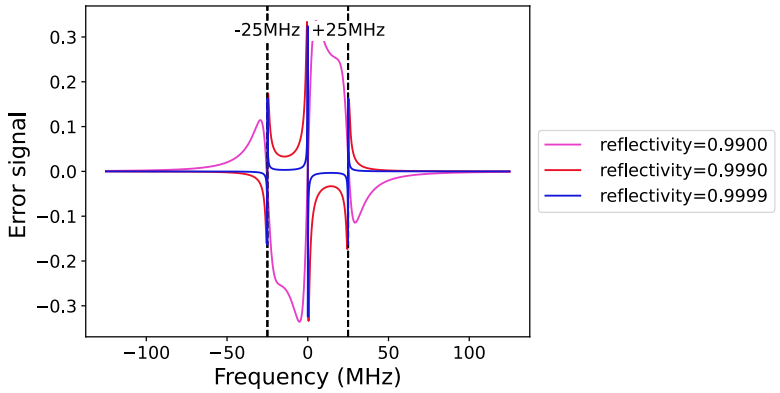


Figure 2.6: We show the PDH error signal for different reflectivities of the cavity mirrors. We identify three peaks, corresponding to the reflection resonance and the two sidebands driven by the EOM at 25 MHz. The signal is linear around the central peak. We increase the reflectivity, and therefore the cavity finesse, which induces sharper peaks.

2.3 PDH Feedback Control and Servo Bandwidth

In this section, we describe the feedback control scheme used to stabilize the laser frequency to a cavity resonance in a PDH lock. After introducing the general structure of the loop, we discuss stability criteria and the concept of

servo bandwidth, which determines the frequency range over which the laser frequency fluctuations are effectively suppressed. The theoretical framework for feedback control presented in this section follows standard references in control theory, in particular Åström and Murray [10] for a general treatment of linear control systems and Bechhoefer [11]. The doctoral thesis of Robin Oswald was also a valuable reference [12].

The basic scheme of the PDH lock can be modeled as a Linear Time Invariant loop control system, with an open-loop transfer function $G_{\mathrm{OL}}(s)$, where $s = j\omega$. The open-loop transfer function is composed of the sensing function $H(s)$, the controller $C(s)$, and the actuator $P(s)$, with an additional time delay τ due to signal propagation and electronics:

$$G_{\mathrm{OL}}(s) = H(s)\, C(s)\, P(s)\, e^{-s\tau}.$$

The sensing function $H(s)$ is the conversion from frequency deviation to error signal (photodiode and PDH electronics), $C(s)$ is the feedback controller and $P(s)$ is the actuator, the physical mechanism that changes the laser frequency in response to the controller output (laser piezo or current modulation).

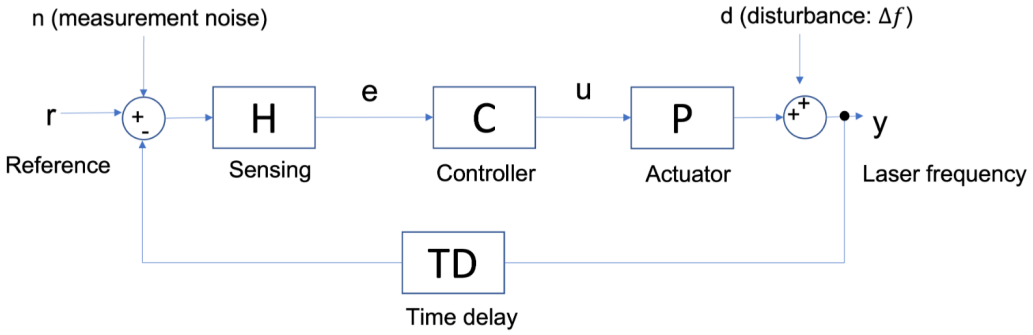


Figure 2.7: Illustration of the closed-loop system where the error signal *e*, coming from laser's frequency and cavity reference comparison, is processed by the PID controller. The controller generates a correction signal *u* that is fed back to the laser's frequency-tuning elements (actuator). *d* represents the actual disturbance on frequency noise that the loop is trying to correct, while *n* is the noise introduced by the detection process. Image from [13].

The closed-loop transfer function is then

$$G_{\mathrm{CL}}(s) = \frac{G_{\mathrm{OL}}(s)}{1 + G_{\mathrm{OL}}(s)},$$

which determines the response of the laser frequency to disturbances, so how well the laser frequency follows the reference. The sensitivity function

$$S(s) = \frac{1}{1 + G_{\mathrm{OL}}(s)}$$

describes how external frequency disturbances are attenuated; for $|G_{\mathrm{OL}}(s)| \gg 1$, disturbances are strongly suppressed.

In practice, we commonly use a proportional-integral (PI) controller. The PI controller transfer function can be written as

$$C(s) = K_P + \frac{K_I}{s},$$

where K_P and K_I are the proportional and integral gains, respectively. More specifically, the proportional term generates a correction signal that is directly proportional to the instantaneous error signal. In the frequency domain, the proportional contribution corresponds to a constant gain, which uniformly amplifies the open-loop transfer function over a broad frequency range. The integral (I) term, on the other hand, integrates the error over time, leading to a gain that increases as $1/\omega$ at low frequencies. This ensures that constant or slowly varying frequency offsets are fully corrected.

A characteristic feature of the closed-loop response is the appearance of *servo bumps* in the sensitivity function and closed-loop function, and thus in the frequency noise spectrum. These instability peaks arise when the open-loop function G_{OL} is close to -1, causing the denominator of the sensitivity function to approach 0. In this condition, the loop does not reject noise, but instead amplifies it. The height of the servo bumps, which can be quantified as the maximum amplitude of the frequency noise above the baseline in the FND spectrum, depends on the choice of K_P and K_I: increasing K_P raises the open-loop gain and pushes the unity-gain frequency into a region with larger phase lag (closer to 180°), thus increasing the resonance peak (servo bump) in the closed-loop response [11]. The location of the servo bumps thus provides a practical estimate of the servo bandwidth, indicating the frequency range over which the loop effectively suppresses disturbances.

Chapter 3

Design and Simulations

3.1 PDH Optical Setup

The aim of the optical setup is to realize the PDH locking scheme as in Fig. 2.4. We had to guide the laser beam through an Electro-Optic Modulator (EOM) that provides the necessary phase modulation, then we sent it to the stable reference cavity. We also had to isolate the reflection signal by the incoming beam and send it to a photodetector to be converted into electronic signal.

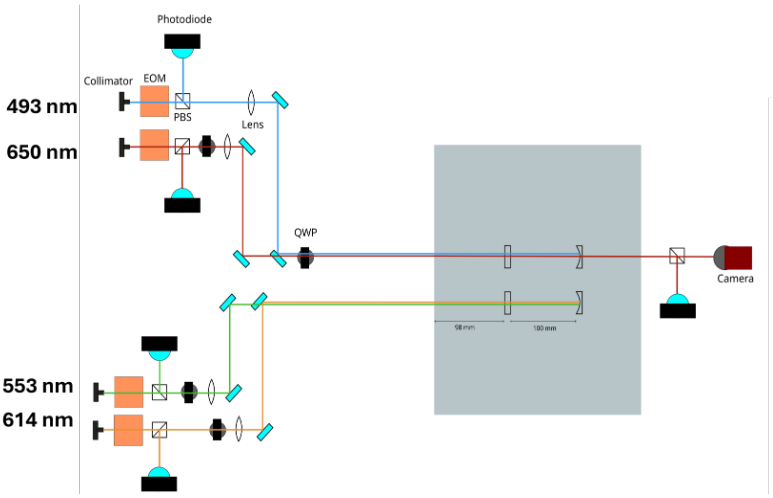


Figure 3.1: All the optical elements after the fiber and before the cavity are positioned on a 450 x 600 mm breadboard, which hosts all the four beam lines. The representation is a projection on the horizontal plane: in the implementation of the setup, the overlapping beams are travelling at different heigths and directed to one cavity each.

After the laser source, we delivered a fraction of the optical beam to the experiment through a polarization-maintaining (PM) fiber, ensuring a stable

and well-defined polarization state. At the fiber output, we used a collimator to convert the divergent beam into a collimated beam, suitable for further manipulation. We then passed the collimated beam through an electro-optic modulator (EOM), where we applied phase modulation via the Pockels effect to generate the sidebands required for the Pound–Drever–Hall stabilization scheme. Subsequently, we employed a mode-matching lens to adapt the beam waist and wavefront curvature: for efficient cavity coupling, the wavefront radius have to match the radius of the cavity mirrors at the corresponding positions. We finally directed the beam into the cavity using a set of steering mirrors. In transmission, we collected the cavity output and split it with a beam splitter, directing the light simultaneously to a camera and a photodiode. The camera provided real-time spatial information about the shape of the transmitted beam for alignment and mode-matching optimization, while the photodiode allowed monitoring of the transmitted optical power during cavity operation.

We controlled the polarization of the beam along the optical path to ensure the separation of the reflected signal from incoming beam. Light exiting the polarization-maintaining fiber was linearly polarized along an axis, and we then directed it through the PBS. One polarization was transmitted straight through, while the orthogonal one was reflected by the PBS at 90°. Before entering the cavity, we passed the beam through a quarter-wave plate (QWP) oriented at 45°, which converted the linear polarization into circular polarization. After reflection from the cavity input mirror, the handedness of the circular polarization was reversed, and after a second pass through the quarter-wave plate, the reflected beam became linearly polarized orthogonally to the incident light. This allowed the PBS to spatially separate the reflected signal from the incoming beam for detection.

3.1.1 Structure of the cavity

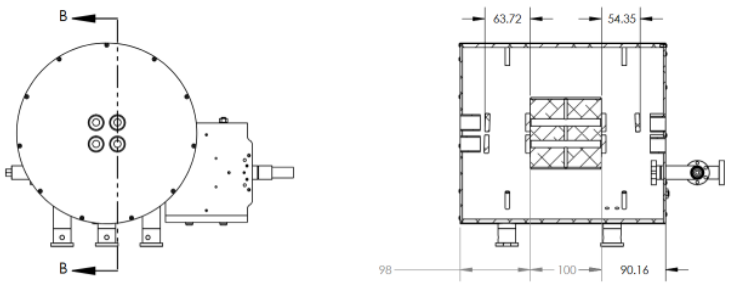


Figure 3.2: CAD drawing of front and lateral view of the four-bore cavity from Stable Laser Systems (SLS). The vertical scale should not be trusted and actual heights have been verified through direct measurement. The SLS datasheet is [14].

The cavity that we used as stable frequency reference for the PDH lock is a low-finesse four-bore cavity from Stable Laser Systems (SLS). It consists of four different sets of mirrors, thus forming four distinct cavities, one per each bore. Each cavity is formed by a planar mirror and a concave mirror with radius of curvature of 50 cm. The curved mirrors are on the front side of the cavity (left side in the lateral view on Fig. 3.2). They are separated by 100 mm nominal distance, corresponding to a Free Spectral Range of 1.5 GHz in vacuum. However, the mirror distance is tunable with a Piezoelectric Transducer (PZT) applied to the concave mirror. The cavity presents four apertures which allow to use the cavity as stable reference for all the four laser beams. The first aperture (lowest corner on the left, looking at Fig. 3.2) has been chosen for 493 nm wavelength.

The cavity is enclosed in a temperature-controlled thermal shield placed inside a vacuum chamber. The vacuum environment suppresses refractive index distortions due to the presence of air, while the active temperature stabilization minimizes thermal drifts of the cavity length. In particular, the spacer material is operated close to its zero-crossing temperature of the coefficient of thermal expansion (CTE), where its length is minimally sensitive to temperature variations. This strongly reduces cavity length fluctuations and therefore improves the frequency stability of the reference.

3.1.2 Gaussian beam simulation

Efficient coupling of the incident laser beam into the optical cavity requires matching of the spatial mode of the incoming field to the target mode of the cavity (usually TEM00). This cavity mode is characterized by a specific beam waist size and wavefront curvature determined by the cavity geometry and mirror radii of curvature. Our goal is to maximize the overlap of the incoming beam to TEM00, so that a clean and stable PDH signal can be achieved. We therefore introduced a mode-matching lens in the optical path to adjust the beam waist position and size of the incoming Gaussian beam such that its spatial profile and curvature coincide with those of the cavity fundamental mode at the input and output mirrors.

Mode-matching is dependent on the wavelength of the incoming beam and its beam size at fiber's output. The waist is set by the fiber numerical aperture (NA). For each wavelength, we computed it from the divergence angle $\theta = \arcsin(\mathrm{NA})$, via:

$$w_0 = \frac{\lambda}{\pi \tan\theta}, \tag{3.1}$$

which corresponds to a beam that emerges from the fiber with its waist located at the output plane.

The beam was propagated through the collimator stage and the mode-matching lens, considering then the free propagation path lens–cavity. The q-parameter at the cavity entrance was obtained via ABCD transformation rule (see Eq. 2.6).

The cavity is then constructed via a sequence of ABCD matrices representing mirror curvatures, substrate propagation and the free-space gap between mirrors. The matrices used for each element are:

- Free-space propagation of length L:

$$M_{\text{prop}} = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

- Thin lens or curved mirror with focal length f (mirror radius R):

$$M_{\text{lens}} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix}, \qquad M_{\text{mirror}} = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R} & 1 \end{pmatrix}$$

From the round-trip ABCD matrix, the calculations retrieve the cavity eigenmode q-parameter by solving the self-consistency condition Eq. 2.6 with $q_{out} = q_{in}$.

We quantified mode matching by evaluating the coupling efficiency between the injected beam and the cavity eigenmode. We did this by computing the analytical expression of the mismatch between the incoming and cavity q-parameters, as given in [15]. The full expression of the coupling efficiency is

$$\eta = \frac{4}{\left(\frac{w_0}{w_p} + \frac{w_p}{w_0}\right)^2 + \left(\frac{\lambda}{\pi w_0 w_p}\right)^2 (z_p - z_0)^2}, \tag{3.2}$$

where w_0 and w_p are the waist size of the cavity mode and of the injected beam, respectively, while z_p and z_0 are the beam waist positions. Finally, we performed a numerical optimization to find a geometrical compromise that maximizes coupling. This used Nelder–Mead simplex optimization (scipy.optimize.minimize [16]), starting from an initial guess and searching within specified parameter bounds. The free parameters that we optimized were the collimator focal length, the focal length of the mode-matching lens, and the lens–cavity distance. We chose focal lengths to match off-the-shelf lenses and collimators from Thorlabs.

Table 3.1: Optimized mode matching parameters.

Wavelength [nm]	Collimator FL [mm]	Lens FL [mm]	Lens–cavity distance [mm]
493	4.6	600	444
553	4.6	600	436
614	4.6	600	425
650	4.6	600	418

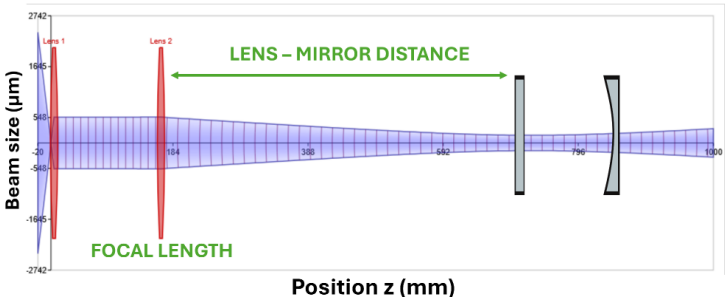


Figure 3.3: Simulation of the incoming Gaussian beam envelope, for 493 nm wavelength. Lens 1 represents the collimator with focal length 4.6 mm, Lens 2 represents the mode matching lens, with focal length 600 mm. The profile of this beam should match the cavity geometry to overlap with the cavity eigenmodes. In this case, the beam waist (where the beam has a planar wavefront) corresponds to the first planar mirror of the cavity, while the beam radius at the concave mirror should be close to 50 cm for a good mode matching.

Chapter 4

Methodology

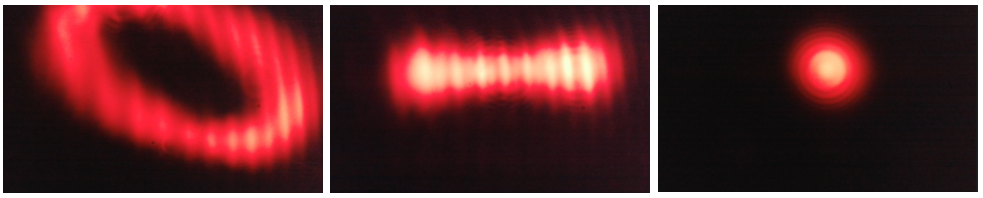
4.1 Optics alignment

Using two steering mirrors, we aligned the laser beam to the cavity axis by carefully controlling the input angle. In parallel, we adjusted the beam height to ensure proper overlap with the cavity optical axis. As a second step in the alignment procedure, we used a pinhole to overlap the incident beam with the light back-reflected from the cavity input mirror, ensuring that the incoming beam was orthogonal to the cavity mirrors.

After this procedure, the transmitted light was visible on the other side of the cavity, while scanning the laser frequency over at least one FSR to ensure that all cavity modes could be observed. The laser frequency was scanned by applying a voltage to a piezoelectric actuator that controls the length of the laser cavity.

Further steps required placing a camera sensitive to the visible spectrum to observe the spatial features of the transmitted signal. We then maximized the coupling by iteratively walking the steering mirrors until the camera image showed a clear, round shape, as shown in Fig 4.1. In practice, walking the beam consists of making a small adjustment to one steering mirror (by turning its adjustment screw) and then compensating with the second mirror. By repeating this a few times horizontally and vertically, the beam alignment converges toward the optimal condition. Importantly, the walking procedure must be performed separately for the horizontal and vertical axes, adjusting only one steering direction at a time to avoid coupling the two degrees of freedom.

Although all transverse modes of a given optical cavity share the same Gaussian beam envelope, characterized by identical waist size, Rayleigh range, and wavefront curvature, efficient coupling into a specific mode is determined by the spatial profile of the incident field. Mode-matching optics



(a) (b) (c)

Figure 4.1: Initially (a), after coarse alignment by eye and by overlapping the back-reflected beam to the incoming beam, the transmission observed on the camera showed an elliptical pattern. Then (b), we walked the mirrors in the vertical direction: this reduced the vertical mismatch and compressed the transmitted pattern into a horizontally elongated shape. After making this pattern as compact as possible, we walked the mirrors in the horizontal direction. This caused the transmission pattern to collapse into a symmetric circular spot (c), corresponding to optimal coupling.

adjust only the global beam parameters, ensuring that the Gaussian envelope of the incoming beam matches that of the cavity modes, but they do not fully control into which transverse mode the beam couples into. Instead, a good alignment and beam quality ensure that the Gaussian input beam closely matches the fundamental TEM_{00} mode, while coupling to higher-order modes is suppressed.

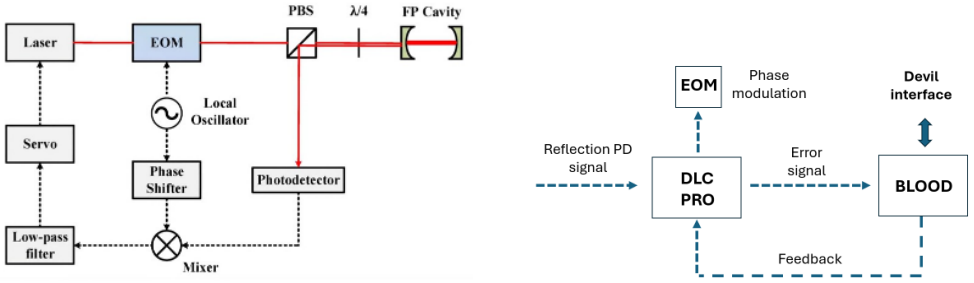
To optimize coupling to the TEM_{00} mode, we observed the cavity transmission or reflection spectrum while scanning the laser frequency. We placed a beamsplitter immediately after the cavity to send the transmitted signal both to the camera and to a photodiode, allowing us to see the transverse mode profiles on the camera and to measure the spectrum on the photodiode. We then further optimized the alignment by adjusting the steering mirrors to minimize the height (intensity) of the peaks corresponding to higher-order transverse modes.

4.2 Electronic and PID Control system

To obtain an effective PDH error signal, we must process the photodetector signal at cavity reflection (see Sec. 3.1), as we describe in Sec. 2.2.1.

Figure 4.2 a) shows the PDH electronics scheme. A local oscillator drove the EOM, which provided phase modulation to the input laser beam. We mixed the same oscillator signal with the cavity reflection signal, while maintaining the ability to adjust the relative phase between the two. After mixing, a low-pass filter generated the desired error signal.

We implemented this electronic path in the actual setup using two main components: the *Digital Laser Controller* (DLC Pro) from TOPTICA, which generated the error signal from the photodetector output, and the *Bichan-*



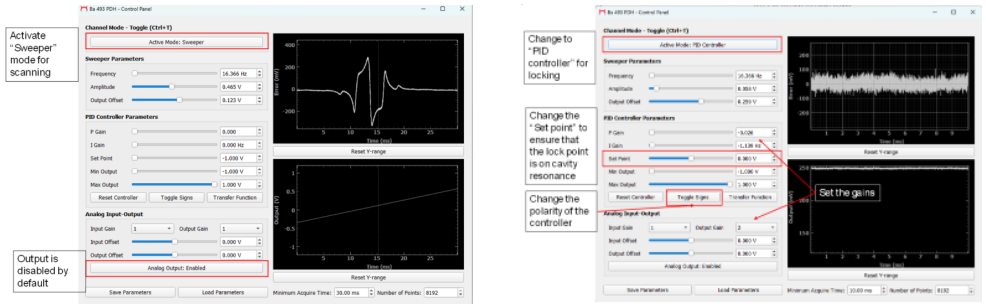
(a) PDH schematics **(b)** Implementation of PDH electronics part

Figure 4.2: (a) Basic electronic scheme showing the local oscillator driving the EOM, the mixing of the reflection signal, and the low-pass filtering to generate the error signal. (b) The physical implementation of the system, where the TOPTICA DLC Pro manages the internal modulation/demodulation and mixing, while the BLOOD PI controller processes the error signal to provide frequency feedback to the laser.

nel Lockbox On One Device (BLOOD), a PI controller that the Trapped Ion Quantum Information Group developed and commonly uses (see [17]).

The DLC Pro integrates the local oscillator and mixer. We adjusted the relative phase between the mixed signals from the control panel (PDH Demodulation Phase) and set the local oscillator frequency (PDH Modulation Frequency) to 25 MHz.

With a driving power of 8.8 dBm, we applied an additional 7 dB of amplification between the oscillator and the EOM, which delivered a total modulation power of 15.8 dBm to the EOM input. It is possible to tune the driving power of the DLC Pro to reach an optimal sideband-to-carrier power ratio near 0.42. This value maximizes the slope of the central region of the PDH error signal (around the carrier zero-crossing), improving the sensitivity to frequency fluctuations.



(a) Devil interface - sweeper **(b)** Devil interface - setting parameters

Figure 4.3: (a) The sweeper mode within the Devil user interface, used to visualize the error signal during frequency scans. (b) The PID configuration panel on Devil interface, which allowed us to manually tune the proportional and integral gains.

The BLOOD received the error signal that the DLC Pro generates as its input. We connected its output to the DLC Pro, which then provides feedback to the laser by adjusting the laser cavity length via the piezo actuator. The limited bandwidth of the BLOOD of approximately 1 MHz acted as the low-pass filter required by the PDH scheme in Fig. 4.2 a).

Fig. 4.3 shows the BLOOD control panel (Devil interface). We used the following procedure for locking the laser frequency:

- While scanning the laser frequency to observe the error signal (sweeper mode), we adjusted the sweeper parameters to center the scan around the carrier frequency, ensuring we stayed only in the linear region of the error signal where the feedback loop works.
- After enabling the lock mode (PID controller), we optimized the P and I parameters. First, set a non-zero value for the I gain. Then, gradually increase the Output Gain until the system locks. Next, alternate between increasing the P and I gains until the lock begins to oscillate. Finally, reduce the gains slightly until the oscillations disappear. For detailed tuning guidelines, refer to the DLC Pro Manual from TOPTICA ([18]).
- Use the toggle sign command to invert the lock polarity. If we select the wrong polarity, the servo locks to one of the sidebands instead of the central slope of the error signal. In fact, the sidebands have an opposite slope compared to the central resonance. If the lock polarity is wrong, what should be a stable negative feedback around the central slope becomes unstable, and the servo tends instead to lock to the sidebands, where, however, the feedback is worse.

Chapter 5

Experimental Results

This chapter presents the results obtained from the data collected during the implementation of the PDH lock, including the cavity characterization and the analysis of the lock stability and performance.

5.1 Cavity Transmission Spectrum, Free Spectral Range and Finesse

The cavity transmission spectrum (see Sec. 4.1), which we acquired while scanning the laser frequency over at least one FSR, appears in Fig. 5.1. For the 650 nm laser, we directly recorded the reflection spectrum, due to our higher confidence in the expected behavior.

The dominant peaks (or dips) correspond to the TEM_{00} cavity mode and repeat every free spectral range. We checked this by centering the scanning window only around the peak and looking at the beam shape on the camera, which resembled TEM00 one in Fig. 2.1. We optimized the beam alignment into the cavity to couple predominantly to the TEM_{00} mode. Nevertheless, some higher-order transverse modes remain visible. The most evident peak corresponds to the TEM_{20} mode, with an extinction ratio of approximately 1/12 for the 493 nm spectrum and 1/20 for the 650 nm spectrum.

We obtained the original data from the TOPTICA DLC Pro GUI, consisting of the voltage we applied to scan the piezo (x-axis) and the corresponding photodetector intensity (y-axis). In the post-processing stage, we rescaled the x axis to convert the applied scanning voltage into the corresponding laser frequency variation. We performed this calibration using the first-order sidebands that the EOM phase modulation generates. We recalibrated the x-axis by measuring the voltage difference between two sidebands around the same carrier, which we expect to correspond to 50 MHz, i.e. twice the EOM modulation frequency. We determined the exact sideband positions

by fitting a Lorentzian function to the data, after subtracting the dominant Lorentzian contribution of the carrier peak. We obtained the final calibration factor by averaging over several carrier–sideband triplets and excluding clear outliers.

After converting the x-axis into frequency units, we can experimentally determine the free spectral range of the cavity from the frequency separation between two adjacent TEM_{00} peaks.

For the 493 nm cavity, we find

$$\mathrm{FSR} = 1504 \pm 4 \text{ MHz},$$

while for the 650 nm cavity,

$$\mathrm{FSR} = 1523 \pm 5 \text{ MHz}.$$

These values show reasonable agreement with the expected free spectral range we derive from the cavity length,

$$\mathrm{FSR} = \frac{c}{2L} = 1499 \text{ MHz}.$$

To obtain a reliable estimate of the cavity finesse, we performed a high-resolution scan around a single carrier peak (dip) and its corresponding sidebands. After calibration, we applied a Lorentzian fit to the carrier peak and extracted the Full Width at Half Maximum from the fit parameters:

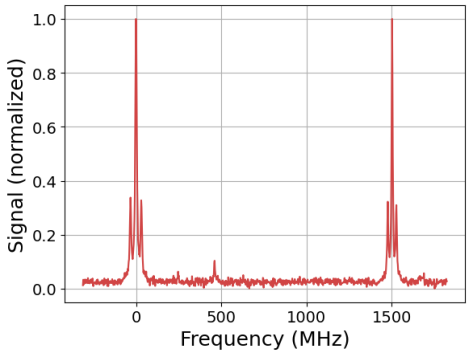
$$\mathrm{FWHM}_{493} = 7.24 \pm 0.10 \text{ MHz},$$
$$\mathrm{FWHM}_{650} = 6.70 \pm 0.02 \text{ MHz}.$$

We then calculate the finesse as the ratio between the experimental FSR and the measured linewidth:

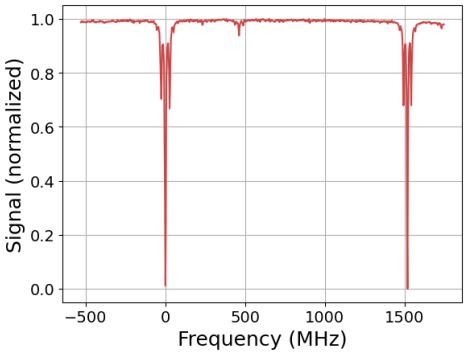
$$\mathcal{F}_{493} = 211 \ \pm \ 5,$$
$$\mathcal{F}_{650} = 227 \ \pm \ 1.$$

5.2 PDH Error Signal Analysis

In order to characterize the cavity and validate the implementation of the Pound–Drever–Hall technique scheme, we looked at the error signal in the output to the DLC Pro, while scanning the laser frequency. The measured signal exhibits the expected antisymmetric shape around resonance (Fig. 5.3), with a zero crossing at the cavity resonance frequency that was chosen to

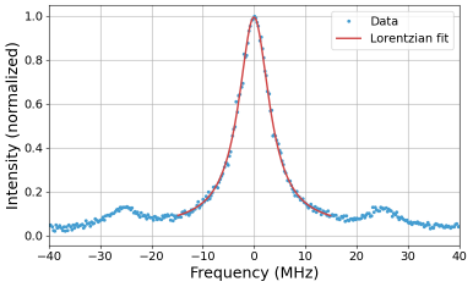


(a) Transmission spectrum of 493 nm laser through the Fabry-Pérot cavity

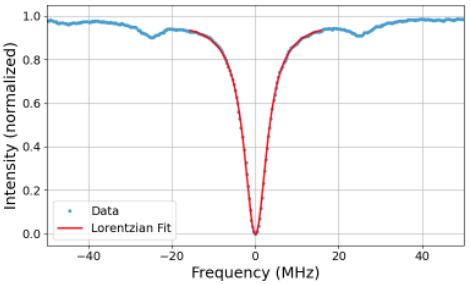


(b) Reflection spectrum of 650 nm laser from the Fabry-Pérot cavity

Figure 5.1: Transmission peaks for the (a) 493 nm and reflection dips for the (b) 650 nm lasers as they are scanned across one cavity Free Spectral Range. The x-axis was rescaled, converting the scan voltage units to frequency shift units, using sideband to carrier known distance of 25 MHz to calibrate. The dominant peaks correspond to two consecutive TEM00 cavity modes. We achieved extinction ratios of $1/12$ for 493 nm and $1/20$ for 650 nm between TEM00 peaks and higher order transverse modes, that are barely visible.



(a) Lorentzian fit for transmission peak of 493 nm laser



(b) Lorentzian fit for reflection dip of 650 nm laser

Figure 5.2

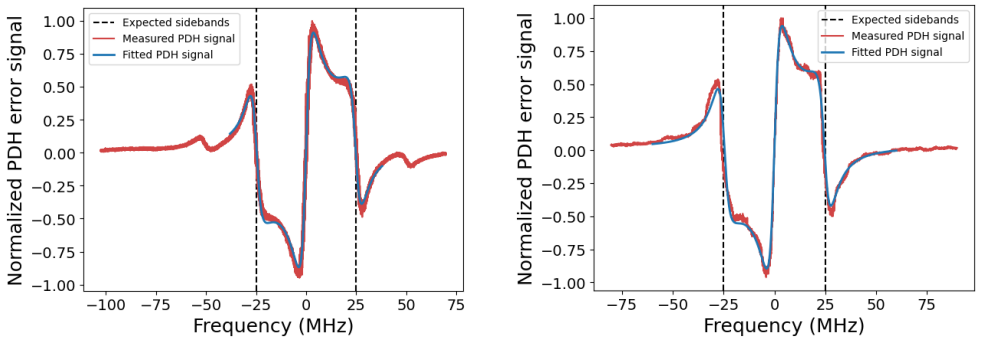
be the locking point, and symmetric side features arising from the phase modulation sidebands.

We fitted the experimental data with a theoretical PDH error signal model derived from the cavity reflection coefficient and from the Jacobi–Anger expansion of the phase-modulated field (see Sec. 2.2.1). The fit shows good agreement with the measured trace over the central scan range and up to the first sidebands position, accurately reproducing both the slope at resonance and the overall lineshape. From the fit parameters, we extracted the mirror reflectivity to determine the cavity Finesse (Eq. 2.16):

$$\mathcal{F}_{\text{fitted},493} = 210 \pm 1$$
$$\mathcal{F}_{\text{fitted},650} = 202 \pm 1$$

The finesse obtained is consistent with the expected value based on the transmission peak linewidth (see Sec. 5.1), confirming the validity of our theoretical model.

Since second-order sidebands are visible in the 493 nm signal at ± 50 MHz, we extended the model to include the ±2 modulation terms. However, the fit quality deteriorated and the second-order sidebands could not be reproduced accurately. A possible reason is a non-ideal phase modulation in the EOM. These effect would modify the relative amplitudes and phases of the sidebands, leading to deviations from the ideal theoretical model.



(a) Fitted PDH error signal - 493 nm

(b) Fitted PDH error signal - 650 nm

Figure 5.3: Experimentally acquired error signal for the (a) 493 nm laser and (b) 650 nm laser, overlaid with a theoretical fit. The signal exhibits the characteristic linear shape at the center resonance, along with the two sidebands generated by the 25 MHz EOM modulation. In (a), second order sidebands are also visible at $\pm$ 50 MHz. However, an attempt to include them in the fitting model decreased the fit accuracy.

5.3 Locked laser performance

To characterize the stability and noise performance of the locked laser, we need to quantify its frequency fluctuations relative to the cavity resonance. The PDH error signal provides direct access to these frequency deviations, as around the carrier zero-crossing point the signal is approximately linear with respect to detuning from the locking point. We therefore performed a linear fit of the central region of the error signal to extract the slope at the locking point. This slope provides the conversion factor between error signal voltage, that we read at DLC Pro output, and frequency deviation, allowing us to convert the measured voltage noise into frequency noise. While the laser is locked, we therefore acquire a time series of voltage deviations, that are converted into frequency deviations.

5.3.1 Frequency noise

To characterize the frequency stability of the locked laser, we analyzed the noise in terms of Power Spectral Density (PSD) and Frequency Noise Density (FND). The PSD describes how the power of frequency fluctuations is distributed over Fourier frequency, while the FND represents the square root of the PSD.

We computed the PSD using the logarithmic power spectral density (LPSD) function implemented on Python, which is particularly suitable for long time series. To properly capture both low- and high-frequency behavior, we analyzed time series of different durations. A low-pass filter at 5 MHz was applied to the photodetector signal to prevent aliasing effects due to the limited noise sampling frequency for long time series.

The FND plots (Fig. 5.4) give an insight into the dominant noise sources. At low Fourier frequencies, excess noise is typically associated with thermal drifts, electronical (from main power supply) and acoustic perturbations. At high frequencies (typically above 1 MHz), the spectrum generally converges to the white noise floor, which is set by electronic noise in the laser and ultimately by shot noise. This white noise level determines the intrinsic linewidth of the laser, i.e. the narrowest linewidth the laser could achieve in the absence of low-frequency noise:

$$\nu_{\text{intr}} = S_0^2 \, \pi,$$

with S_0^2 the white noise floor of the FND. However, the effective linewidth measured in practice is generally broader, as it also includes contributions from low-frequency noise [19].

In the FND plots, we also show the so-called β-line. The β-line separation method [20] provides a practical criterion for estimating the laser linewidth from the frequency noise spectrum. It defines a frequency-dependent threshold: Noise contributions above the β-line effectively broaden the central linewidth, while contributions below it mainly affect the spectral wings. The intersection between the FND curve and the β-line therefore allows an estimate of the effective linewidth of the laser, by integrating the Frequency noise above the crossing point, at low frequencies:

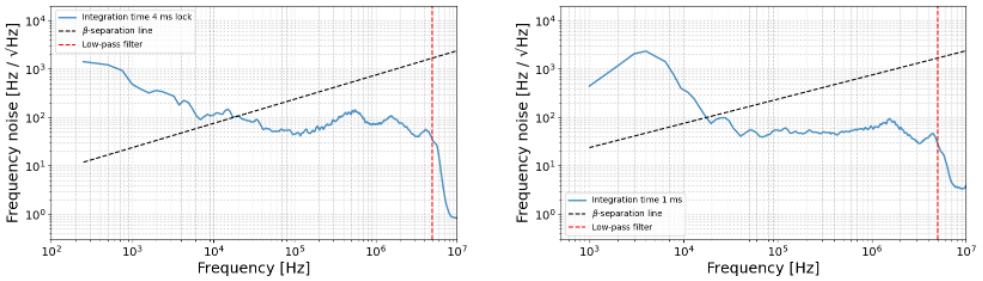
$$\Delta\nu_{\text{eff}} \approx \sqrt{8 \ln 2 \int H\Big([S_\nu(f)]^2 - \frac{8 \ln 2}{\pi^2} f\Big) \, [S_\nu(f)]^2 \, df}, \tag{5.1}$$

where $S_\nu(f)$ is the PSD of the signal at Fourier frequency f and $H(\cdot)$ is the Heaviside function.

We estimated the locked laser linewidth for both 493 nm and 650 nm lasers considering a 4 ms time series for 493 nm and 1 ms series for 650 nm. For the

493 nm laser, the effective linewidth is 74 kHz , while for 650 nm is 329 kHz. The linewidth for 493 nm laser is comparable with fabrication data of the unlocked laser provided by the company. The 650 nm linewidth is higher, as is also possible to see from the FND plot. An explanation for this could lie in the characteristics of the laser itself or could be determined by non optimal PI lock parameters. The reasons should be investigated further.

Note that β-separation gives an effective linewidth, but if low-frequency noise dominates and laser frequency drift is present, then the result becomes strongly dependent on measurement time. Longer scans present a lower integration bound and therefore the reconstructed linewidth is increased.



(a) Frequency Noise Density - 493 nm **(b)** Frequency Noise Density - 650 nm

Figure 5.4: The frequency noise density is shown for the (a) 493 nm and (b) 650 nm lasers while locked to the multi-bore cavity in a PDH loop. The red vertical line represents a low-pass filter at 5 MHz, to avoid aliasing of data. The dashed black line represents the β-separation line, which serves as a boundary to distinguish between noise that contributes to the laser linewidth at low frequencies and noise that only affects the lineshape wings. The 650 nm lock presents more intense residual noise across the spectrum compared to the 493 nm system. This residual noise in the 650 nm line indicates a broader effective linewidth and less efficient noise cancellation.

5.3.2 Servo bump measurement

To characterize the lock performance, we aimed to estimate the PDH lock bandwidth in order to determine the frequency range over which the feedback loop actively suppresses noise.

As discussed in Sec. 2.3, one method to estimate the bandwidth is to identify servo bumps, which appear as peaks in the Frequency Noise Density. Their amplitude changes when the P and I gains of the lock are varied.

The results are shown in Fig. 5.5, where several time series of equal duration were recorded while progressively increasing the P gain. Servo bumps are clearly visible around 500 kHz. The PDH loop becomes unstable at high frequencies when the P parameter is set to a sufficiently high value (e.g. $P = 1$ or larger). From these observations, the bandwidth of the PDH lock can be estimated to be slightly below 500 kHz.

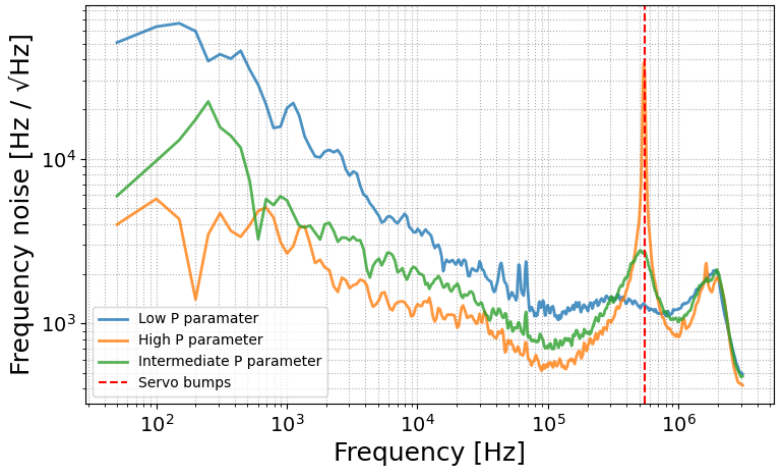


Figure 5.5: A series of frequency noise density spectra recorded with increasing proportional (P) gain for the 493 nm laser. The emergence of servo bumps near 500 kHz indicates the frequency limit (bandwidth of the system) where the PDH loop loses stability and amplifies the frequency noise, instead of reducing it.

Chapter 6

Conclusions and Outlook

In this project, we successfully designed, implemented, and characterized a Pound–Drever–Hall (PDH) locking setup to stabilize the frequencies of the 493 nm and 650 nm laser lines to a shared multi-bore reference cavity. Precise laser frequency stabilization is essential for high-resolution spectroscopy and coherent control of trapped ions, and therefore constitutes a key requirement for high-precision atomic parity violation (APV) experiments using trapped $^{138}Ba^{+}$ ions.

We constructed a complete optical and electronic PDH infrastructure using a four-bore cavity from Stable Laser Systems with a free spectral range (FSR) of 1.5 GHz. We placed particular emphasis on efficient coupling to the cavity fundamental mode. Through Gaussian beam simulations, optimization of the mode-matching optics, and alignment efforts, we achieved high coupling efficiency to the TEM_{00} mode, with strong suppression of higher-order transverse modes for both wavelengths.

We experimentally characterized the spectral properties of the cavity by analyzing transmission and reflection spectra. From these measurements, we determined the finesse to be 207 ± 3 for 493 nm cavity and 223 ± 1 for 650 nm. We demonstrated stable frequency locks of both the cooling and repumper lasers using TOPTICA DLC Pro controllers in combination with the BLOOD PI system. From the observation of servo bumps in the frequency noise spectrum, we estimated the lock bandwidth to be slightly below 500 kHz.

As an outlook, the same stabilization scheme will be extended to additional laser beams required in the experiment (614 nm and 553 nm).

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